General Topics
&
Mole Concept

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Basic of Chemistry

INTRODUCTION

There are a large number of objects around us which we can see and feel.

Anything that occupies space and has mass is called matter. Ancient Indian and Greek Philosopher's believed that the wide variety of object around us are made from combination of five basic elements: Earth, Fire, Water, Air and Sky. The Indian Philosopher Kanad (600 BC) was of the view that matter was composed of very small, indivisible particle called "parmanus".

Ancient Greek Philosophers also believed that all matter was composed of tiny building blocks which were hard and indivisible.

The Greek philosopher Democritus named these building blocks as atoms, meaning indivisible. All these people have their philosophical view about matter, they were never put to experimental tests, nor ever explain any scientific truth.

It was John Dalton who firstly developed a theory on the structure of matter, later on which is known as Dalton's atomic theory.

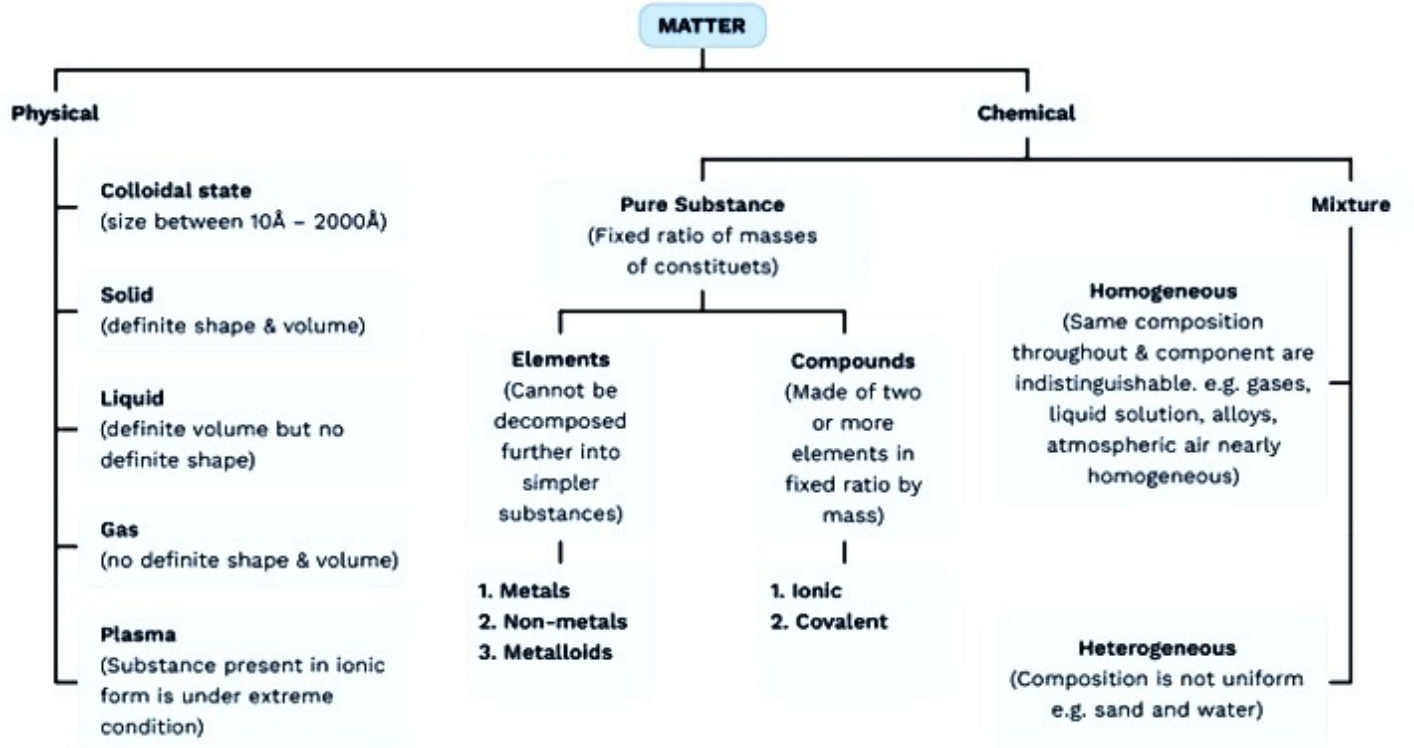
DALTON'S ATOMIC THEORY

Matter is made up of very small indivisible particles called atoms.

All the atoms of a given element are identical in all respect i.e. mass, shape, size, etc.

Atoms cannot be created or destroyed by any chemical process.

Atoms of different elements are different in nature.



LAWS OF CHEMICAL COMBINATION:

Antoine Lavoisier, John Dalton and other scientists formulate certain laws concerning the composition of matter and chemical reactions. These laws are known as the law of chemical combination.

THE LAW OF CONSERVATION OF MASS:

It is given by Antoine Lavoisier.

In a chemical change total mass remains conserved that is mass before reaction is always equal to mass after reaction.

Example:

$H_2(g) + y_2O_2(g) \rightarrow H_2O(l)$

Before reaction	1 mole	y_2 mole	
After the reaction	0	0	1 mole

mass before reaction = mass of 1 mole H_2 (g) + mass of $\frac{1}{2}$ mole O_2 (g)

= 2 + 16 = 18 g

mass after reaction = mass of 1 mole water = 18 g

LAW OF CONSTANT OR DEFINITE PROPORTION:

It is given by Proust. All chemical compounds are found to have constant composition irrespective of their method of preparation or sources.

Example: In water (H_2O), Hydrogen and Oxygen combine in 2 : 1 molar ratio, the ratio remains constant whether it is tap water, river water or sea water or produced by any chemical reaction.

Example : 1.80 g of a certain metal burnt in oxygen gave 3.0 g of its oxide. 1.50 g of the same metal heated insteam gave 2.50 g of its oxide. Show that these results illustrate the law of constant proportion.

Sol. In the first sample of the oxide,

wt. of metal = 1.80 g,

wt. of oxygen = (3.0 – 1.80) g = 1.2 g

$$\frac{\text{wt.of metal}}{\text{wt. of oxygen}} = \frac{1.80\text{g}}{1.2\text{g}} = 1.5$$

In the second sample of the oxide,

$$\frac{\text{wt.of metal}}{\text{wt. of oxygen}} = \frac{1.50\text{g}}{1\text{g}} = 1.5$$

Thus, in both samples of the oxide the proportions of the weights of the metal and oxygen are fixed. Hence, the results follow the law of constant proportion.

THE LAW OF MULTIPLE PROPORTIONS:

It is given by Dalton.

When one element combines with the other element to form two or more different compounds, the mass of one element, which combines with a constant mass of the other, bear a simple ratio to one another.

Note: Simple ratio here means the ratio between small natural numbers, such as 1: 1, 1: 2, 1: 3, Later on this simple ratio becomes the valency and then oxidation state of the element.

Example: Carbon and Oxygen when combine, can form two oxides, CO (carbon monoxide), CO_2(Carbon dioxides)

In CO, 12 g carbon combined with 16 g of oxygen.

In CO_2, 12 g carbon combined with 32 g of oxygen.

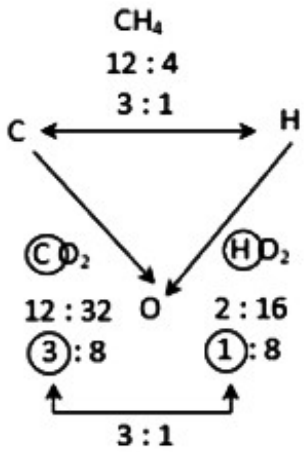
Thus, we can see the mass of oxygen which combine with a constant mass of carbon (12 g) bear simple ratio of 16 : 32 or 1 : 2.

LAW OF RECIPROCAL PROPORTION:

It is given by Richter.

The ratio of the weights of two elements A and B which combine separately with a fixed weight of the third element C is either the same or simple ratio of the weights in which A and B combine directly with each other.

Example:



Example: Ammonia contains 82.35% of nitrogen and 17.65% of hydrogen. Water contains 88.90% of oxygen and11.10% of hydrogen. Nitrogen trioxide contains 63.15% of oxygen and 36.85% of nitrogen. Show that these data illustrate the law of reciprocal proportions.

Sol. In NH3, 17.65g of H combine with N = 82.35g

1 g of H combine with $N = \frac{82.35}{17.65} g = 4.67g$

In H2O, 11.10 g of H combine with O = 88.90 g

1 g of H combine with $O = \frac{88.90}{11.10} g = 8.01g$

Ratio of the weights of N and O which combine with fixed weight (=1g) of H = 4.67 : 8.01 = 1 : 1.7

In N2O3, ratio of weights of N and O which combine with each other =36.85 : 63.15 = 1 : 1.7

GAY-LUSSAC'S LAW OF COMBINING VOLUME:

According to him elements combine in a simple ratio of atoms, gases combine in a simple ratio of their volumes provided all measurements should be done in the same temperature

$$\underset{1\,vol}{H_2(g)} + \underset{1\,vol}{Cl_2(g)} \longrightarrow \underset{2\,vol}{2HCl}$$

AVOGADRO'S H YPOTHESIS:

Equal volumes of polyatomic all gases have equal number of molecules (not atoms) at same temperature and pressures conditions.

S.T.P. (Standard Temperature and Pressure)

At S.T.P. condition:

Temperature = 0°C or 273 K

Pressure = 1 bar

and volume of one mole of gas at STP is found to be equal to 22.7litres which is known as molar volume.

Mole Concept

MOLE

One mole of pure substance contains as many unit particles as there are number of atoms in exactly 12 grams of pure C^{12} sample.

1 mole substance = 6.02×10^{23} unit particle of that element

$6.02 \times 10^{23} = N_A$ (Avogadro's Number)

i.e.

→ 1 mole Ca	= 6.02×10^{23} atoms of Ca
→ 1 mole H_2	= 6.02×10^{23} molecules of H_2
	= $2(6.02 \times 10^{23})$ atoms of H
→ 1 mole O_3	= N_A molecules of O_3
	= 3 N_A atoms of O
→ 1 mole water	= N_A molecules of H_2O
	= $2N_A$ atoms of H
	= N_A atoms of O
	= $3N_A$ total atoms
1 mole substance	= N_A unit particles
For a substance	

Avogadro's Number

- 6.02×10^{23} is Avogadro's Number
- Equivalent of One mole

$$\text{Moles} = \frac{\text{No. of unit particles}}{N_A}$$

Or moles × N_A = No. Of unit particles

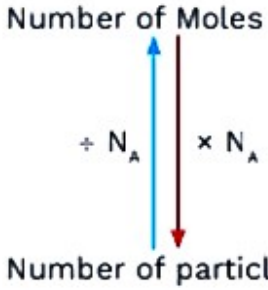


ILLUSTRATION:

Q1 **From one mole sample of iron if 12.04×10^{22} atoms are removed then remaining number of moles will be**

(A) 0.2 **(B) 0.8**

(C) 0.4 **(D) 0.6**

A1 **(B)**

Given, one mole sample of iron

Number of atoms removed from 1 mole sample of Fe = 12.04×10^{22} atoms

Remaining number of moles

= 1 mole iron $-(12.04 \times 10^{22}$ atom)

$$= 1 \text{ mole} - \frac{12.04 \times 10^{22}}{6.02 \times 10^{23}} \text{moles} \qquad \left(\because \text{moles} = \left(\frac{\text{No. of unit particle}}{N_A}\right)\right.$$

= 1-0.2

= 0.8 mole

Q2 **The total number of atoms in 5 moles of $C_6H_{12}O_6$ will**

(A) 60 **(B) 24 N_A**

(C) 120 N_A **(D) 24**

A2 **(C)**

Given, 5 moles of $C_6H_{12}O_6$ = $5N_A$ molecule of $C_6H_{12}O_6$

= $24 \times 5N_A$ total atoms

= 120 N_A

Q3 **In 2.5 g molecule sample of CH_3COOH calculate**

(i) Total number of molecule

(ii) Gram-atom of hydrogen

(iii) Total no of atoms

A3 Given:

2.5 gm-molecule sample of CH_3COOH means 2.5 moles of CH_3COOH

(i) 2.5 moles of CH_3COOH $= 2.5 \times N_A$ molecules

(ii) Gram atom of hydrogen = moles of H

$= 4 \times 2.5$ mole $= 10$ mole

(iii) total no. Of atoms $= 2.5\ N_A \times 8 = 20\ N_A$

Q4 **The magnitude of charge present (in coulombs) in 5 g–ions of $(CO_3)^{2-}$ will be: ($N_A = 6 \times 10^{23}$)**

(A) 2 **(B) 9.6×10^5**

(C) 1.6×10^{-18} **(D) 10**

A4 **(B)**

Given:

5 g ions of $(CO_3)^{2-}$ means 5 moles of CO_3^{2-} ions

5 mole of ions CO_3^{2-} $= 5N_A$ ions of CO_3^{2-}

On one ion, charge $= 2.e^-$ charge

$= 2 \times 1.6 \times 10^{-19}$C

Total charge $= 5N_A \times 2 \times 1.6 \times 10^{-19}$

$= 5 \times 6 \times 10^{23} \times 2 \times 1.6 \times 10^{-19}$

$= 9.6 \times 10^5$

MOLAR MASS

It is defined as the mass of mole substance.
i.e.

Molar mass = Mass of N_A unit particles of substance
= Atomic mass or molecular mass in grams

e.g.

i) **He (atomic mass = 4)**

⇒ Mass of 1 mole He = 4 gram
Or molar mass of He = 4 gram mol^{-1}

ii) **O_3 (molecular mass = 48)**

Mass of 1 mole ozone = 48 g
Molar mass of ozone = 48 g mol^{-1}

iii) **NH_3**

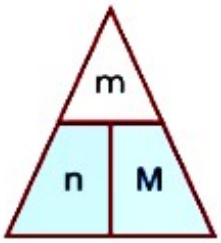
Mass of 1 mole NH_3 or ammonia = 17 gm
Molar mass of ammonia = 17 g mol^{-1}

i.e. M gram substance = 1 mole substance

For substance,

$$\text{Moles} = \frac{\text{Mass of substance(g)}}{\text{Molar mass}}$$

Mass of substance = moles × molar mass



m = n×M

m = Mass
n = Number of moles
M = Molar mass

ILLUSTRATIONS:

Q1 **If mass of 2 atoms of an element is 4×10^{-23} g then atomic mass of element will be: [take $N_A = 6 \times 10^{23}$]**

(A) 24 **(B) 12**

(C) 6 **(D) 2**

A1 **(B)**

Given:

Mass of 2 atoms of an element = 4×10^{-23} g

Mass of 1 mole of an element or mass of N_A atoms $= \frac{4 \times 10^{-23}}{2} \times N_A$

$= \frac{4 \times 10^{-23}}{2} \times 6 \times 10^{23} = 12\,g$

i.e., Molar mass = 12g/mol

Atomic mass = 12

Q2 **The number of carbon atoms present in a signature written by carbon pencil weighs 1.2×10^{-3} gm is**

(A) 12.04×10^{20} **(B) 6.02×10^{19}**

(C) 3.01×10^{19} **(D) 6.02×10^{20}**

A2 **(B)**

Given:

Weight of carbon pencil = 1.2×10^{-3} gm

Moles of carbon = $\frac{\text{Mass of carbon pencil}}{\text{Molar mass}} = \frac{1.2 \times 10^{-3}}{12}\text{mole} = 10^{-4}$ mole

Number of carbon atoms $= 10^{-4} \times N_A$ atoms

$= 10^{-4} \times 6.02 \times 10^{23}$

$= 6.02 \times 10^{19}$

Q3 **The mass of nitrogen present in 3 grams of urea is**

(A) 2 grams **(B) 0.7 grams**

(C) 2.8 grams **(D) 1.4 grams**

A3 **(D)**

Given,

Mass of urea (NH_2CONH_2) = 3 grams

Moles of urea $= \frac{\text{mass}}{\text{molar mass}} = \frac{3}{60} = \frac{1}{20}\text{mole}$

Moles of N–atom $= 2 \times \frac{1}{20}\text{mole} = \frac{1}{10}\text{mole}$

Mass of N–atom = moles × molar mass

$= \frac{1}{10} \times 14\,g = 1.4\,g$

IDEAL GAS

The gas which obey following equation:

$PV = nRT$

Where P = Pressure of gas (atm)

V = Volume of gas contains (L)

R = Universal gas constant $\left(\frac{0.0821\ \text{L atm}}{\text{mole.K}}\right)$

T = Temperature (K)

n = moles of gas

Other forms

$PM = dRT$

Where M = molar mass of gas (g/mol)

d= density of gas (g/L)

$P = cRT$

Where 'c' is concentration of gas (mol/L)

Temperature	RT
0°	24.4
25°C	24.46
27°C	24.63
400K	32.8

ILLUSTRATIONS:

Q1 **The pressure (atm) exerted by 32 grams of $CH_4(g)$ present in 8.21 dm^3 vessel at 127°will be**

(A) 2.0 **(B) 4.0**

(C) 6.0 **(D) 8.0**

A1 **(D)**

Mass of CH_4 = 32 grams

Moles of CH_4 = $\frac{\text{Mass}}{\text{molar mass}} = \frac{32}{16}$ = 2 moles

Volume of vessel = 8.21 dm^3 = 8.21 L

By using ideal gas equation,

$PV = nRT$

$P = \frac{nRT}{V}$

Pressure exerted by CH_4, $P = \frac{2 \times 0.0821 \times 400}{8.21}$ = 8 atm

Q2 **62 g of gaseous phosphorus exerts a pressure of 0.5 atm in 22.4 L vessel at 0°C. The atomicity of phosphorus is**

(A) 1 **(B) 2**

(C) 3 **(D) 4**

A2 **(D)**

Mass of gaseous phosphorus = 62 g

Let phosphorus exist in P_x form.

Molar mass of phosphorus = 31 x g/mol

Pressure = 0.5 atm

Volume of vessel = 22.4 L

Temperature = 0°C = 273 K

Moles of gaseous phosphorous, $n = \frac{62}{31x} \times 0.0821 \times 273$

Using, $PV = nRT$

$0.5 \times 22.4 = \frac{62}{31x} \times 0.0821 \times 273$

$x = \frac{62}{31} \times \frac{0.821 \times 273}{0.5 \times 22.4}$

$x = \frac{2 \times 22.4}{0.5 \times 22.4} = 4$

STP CONDITIONS

STP: Standard temperature and pressure

Old convention	New convention
P=1 atm	**P = 1 bar**
T = 0°C	**T=0°C**

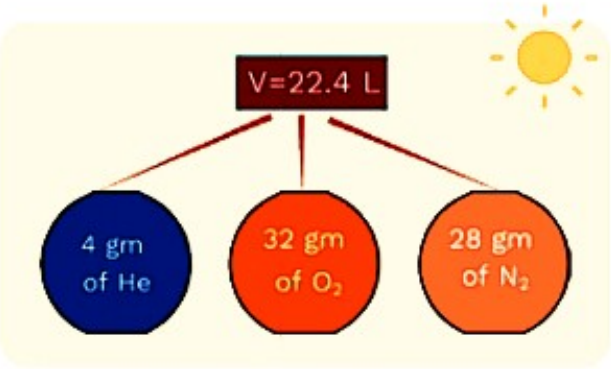
Volume of 1 mole gas at STP

→ According to old convention,

$$PV = nRT$$

$$(1atm)\times V = 1\times 22.4$$

$$V = 22.4\ L$$



→ According to new convention,

P = 1bar

$$= \frac{1}{1.01325}\ atm$$

$$PV = nRT$$

$$\left(\frac{1}{1.01325}\right)\times V = 1\times 22.4$$

$$V = 22.7\ L$$

If 'V' be the volume of 'n' mole ideal gas at STP, then

$V = n\times 22.4\ L$ (Old convention) and $V = n\times 22.7\ L$ (New Convention)

ILLUSTRATIONS:

Q1 **The volume of one mole gas at STP will be**

(A) 5.6 L **(B) 11.2 L**

(C) 22.4 L **(D) 44.8 L**

A1 **(C)**

Fact based –(Old convention based)

Q2 **Which of the following gas sample has maximum number of atoms?**

(A) 44.8 L of O_3 at STP

(B) 49.2 L of H_2 at 1 atm, 300 K

(C) 22.4 L of O_2 at 2 atm, 546 K

(D) 11.2 L of N_2O at STP

A2 **(A)**

Using $PV = nRT$ or $n = \frac{PV}{RT}$

(A) $n_{O_3} = \frac{1 \times 44.8}{22.4} = 2$ mole

No of atoms $= 2 \times 3 \times N_A = 6N_A$

(B) $n_{H_2} = \frac{49.26 \times 1}{24.63} = 2$ mole

Atoms $= 2 \times 2 \times N_A = 4N_A$

(C) $n_{O_2} = \frac{2 \times 22.4}{44.8} = 1$ mole

Atoms $= 1 \times 2 \times N_A = 2N_A$

(D) $n_{N_2O} = \frac{11.2}{22.4} = 0.5$ mole

Atoms $= 0.5 \times 3 = 1.5$

$\therefore$ O_3 has maximum no of atoms

Q3 **The density of a gas at STP is 2/11.35 (grams/L). The gas is**

(A) NH_3 **(B)** Cl_2

(C) SO_2 **(D)** **He**

A3 **(D)**

Using, PM = dRT

$$\frac{1}{1.01325} \times M = \frac{2}{11.35} \times 22.4$$

M = 4 g / mol (approx.)

VAPOUR DENSITY

- Defined only for gases as following:

$$\text{(V.D) gas} = \left(\frac{\textbf{density of gas}}{\textbf{density of } H_2}\right) \textbf{at same T \& P}$$

- Vapour density is a unitless quantity
- For gases, PM=dRT $\Rightarrow d = \frac{PM}{RT}$

$$\left(\text{Vapour density}\right)_{gas} = \frac{\left(\frac{PM}{RT}\right)_{gas}}{\left(\frac{PM}{RT}\right)_{H_2}} = \frac{(M)_{gas}}{(M)_{H_2}}$$

$$\text{Vapour Density} = \frac{\text{Molecular Mass}}{2}$$

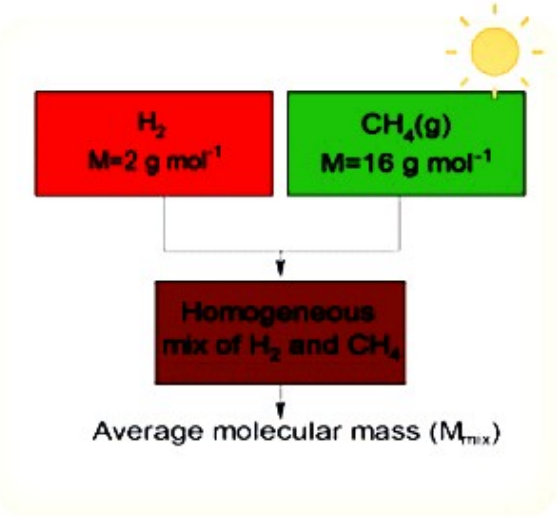
$$\left[\therefore (M)_{H_2} = 2\,gm / mol\right]$$

(V.D)gas× 2 = Molar Mass of Gas (g/mol)

Average Molecular Mass (M_{mix})

It is defined for a mixture of two or more non reacting gases (always homogeneous)

It is also **known as molar mass of gaseous mixture**



We know that

$$n = \frac{W}{M} \text{ or moles} = \frac{\text{mass(in gm)}}{\text{molar mass(in gm / mole)}}$$

$$\text{or } M = \frac{W}{n}$$

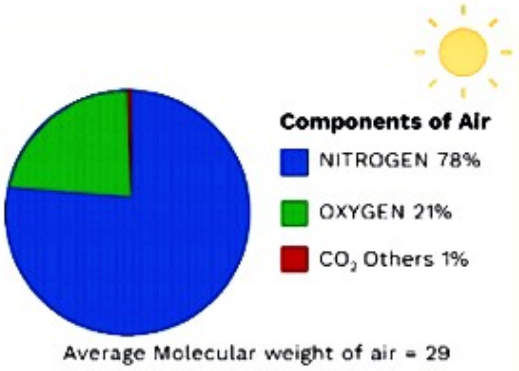
Molar mass of mixture

$$M_{mix} = \frac{\text{Total mass of mixture}}{\text{Total moles in mixture}}$$

Unit of M_{mix} is g mol^{-1}

Where $W_{Total} = \Sigma W$

$n_{Total} = \Sigma n$



Average Molecular weight of air = 29

ILLUSTRATIONS:

Q1 **A mixture contains 1 mole He(g), 2 mole H_2(g) and 2 moles of CH_4(g). The molecular mass of gaseous mixture will be:**

A1 Given, 1 mole He(g), 2 mole H_2 (g), 2 moles CH_4 (g)

n_{mix} = 1+2+2=5 moles

W_{mix} = 1(4) + 2(2) + 2(16)

= 4+4+32 = 40

$$M_{mix} = \frac{W_{mix}}{n_{mix}} = \frac{40}{5} = 8\,g/mol$$

Q2 **A gaseous mixture of Ne and C_2H_6 contains 40% neon by mass. The vapour density of the mixture will be**

A2 Given,

A mixture of Ne and C_2H_6 contains 40% neon by mass

Let total mass = 100 gm

Mass of neon = 40 g

$$\text{moles of Ne} = \frac{40}{20} = 2$$

$$\text{Mass of } C_2H_6 = (100-40)\text{ gm moles } C_2H_6 = \frac{60}{30} = 2$$

$$\text{Molar mass of mixture} = \frac{\text{mass of mixture}}{\text{moles of mixture}}$$

$$\text{or } M_{mix}\ \frac{W_{mix}}{n_{mix}} = \frac{100}{2+2} = 25\ g/mol$$

$$(\text{Vapour density})_{mix} = \frac{M_{mix}}{2} = \frac{25}{2} = 12.5$$

Q3 **The percentage by mole of NO_2 in a mixture of NO_2(g) and NO(g) having average molecular mass 34 is**

(A) 25% **(B) 20%**

(C) 40% **(D) 75%**

A3 **(A)**

Given, Average molecular mass = 34

Let total moles of mixture 100 mole

moles of NO_2 = x mole

moles of NO = (100-x) mole

$$M_{mix} = \frac{W_{mix}}{n_{mix}}$$

$$34 = \frac{(x)\ 46 + (100 - x)30}{100}$$

x=25 mole

AVERAGE ATOMIC MASS (M_{AVG})

Defined for element which have two or more natural isotopes as well as their % abundance or natural occurrence are comparable.

e.g.

i) Hydrogen has two natural isotopes i.e. Protium $\left({}_{1}^{1}H\right)$ and deuterium $\left({}_{1}^{2}H\right)$

Isotopes	${}_{1}^{1}H$	${}_{1}^{2}H$
Abundance	99.99%	0.01%

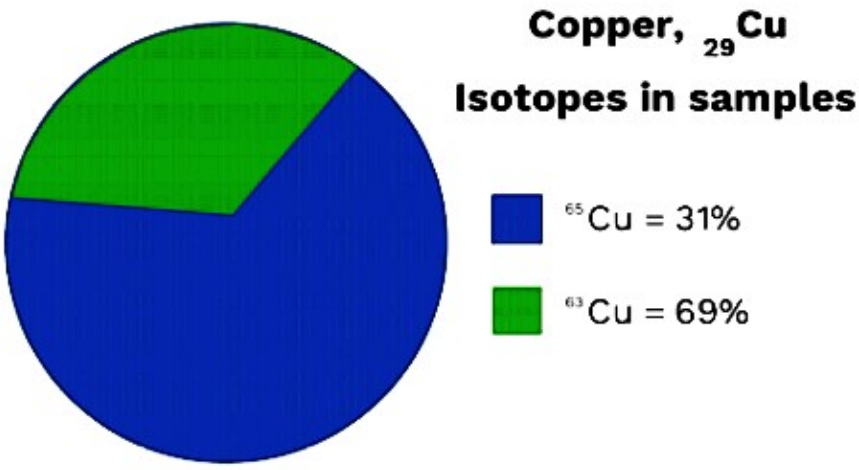
Averge Molar Mass of Hydrogen ≈ 1g / mol

ii) Chlorine has two isotopes

Isotopes	^{35}Cl	^{37}Cl
Abundance	~75%	~25%

Averge Molar Mass of Chlorine (Average) ≈ 35.5 / mol

For such element ,



Average Atomic weight of Cu = 63.55

$$\text{Average atomic mass}(M_{avg}) = \frac{\text{sum of atomic masses of all atoms}}{\text{Total number of atoms}}$$

$$\text{or } M_{avg} = \frac{W_{total}}{n_{total}} \text{ g / mol}$$

ILLUSTRATIONS:

Q1 **In nature Boron exists as ^{10}B and ^{11}B with percent abundance of 40% and 60% by number. The atomic mass of boron will be**

A1 Sol. Given, Boron exists as ^{10}B and ^{11}B
% abundance of ^{10}B = 40%
% abundance of ^{11}B= 60%
Let total number of atoms = 100
No. of atoms of ^{10}B = 40
No. of ^{11}B = 60

$$M_{avg} = \frac{\sum \text{ at masses}}{\text{no. of atoms}}$$

$$M_{avg} = \frac{40(10) + 60(11)}{100} = 10.6$$

Q2 **Chlorine has two naturally occurring isotopes out of them ^{35}Cl is 75% by moles. If atomic mass of chlorine is 35.5 then determine mass number of other isotopes of chlorine.**

A2 Given, percent by mole of ^{35}Cl = 75%
Atomic mass of chlorine = 35.5
Let total no. of atoms =100
No. of atoms of ^{35}Cl = 75
No. of atoms of other isotope = 25

$$M_{avg} = \frac{W_{total}}{n_{total}}$$

$$35.5 = \frac{75(35) + 25(A)}{100}$$

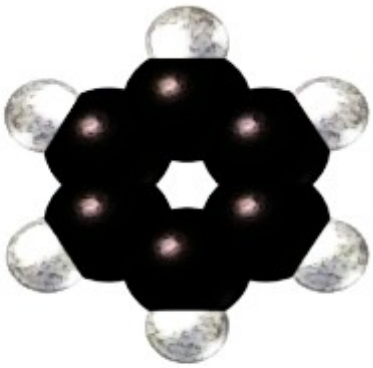
35.5×100=2625+25A
3550-2625 = 25A
925 = 25 A

$$\Rightarrow \quad A = \frac{925}{25} = 37$$

Molecular Formula of Compound

The formula showing actual number of atoms of all elements present in one molecule of compound.

EMPIRICAL FORMULA OF COMPOUND



Benzene

Molecular Formula C_6H_6

Empirical Formula CH

The formula showing simplest ratio of atoms of all elements present in one molecule of compound.

Molecular formula=(Empirical formula)×n

Where 'n' is a positive integer

Determination of M.F

1. From given information, calculate moles of atoms of all elements.
2. Determine simplest ratio of these moles and write down empirical formula.
3. For M.F, find 'n' as following:

$$n = \frac{\text{molecular mass of compound}}{\text{Empirical formula mass}}$$

ILLUSTRATIONS:

Q1 **6.4 grams of a compound of hydrogen and nitrogen contains 0.8 grams of hydrogen. Determine empirical formula of compound.**

A1 Given, mass of hydrogen = 0.8 grams

Mass of compound = 6.4 grams

Mass of nitrogen = (6.4-0.8) g = 5.6 g

$$\text{Moles of hydrogen} = \frac{0.8}{1}\text{mole} = 0.8\,\text{mole}$$

$$\text{Moles of nitrogen} = \frac{5.6}{14}\text{mole} = 0.4\ \text{mole}$$

Simplest ratio (N:H) = 1:2

So, E.F = NH_2

Q2 **If molar mass of compound in previous question is 32g/mol then determine molecular formula.**

A2 N_2H_4

Given, molar mass of compound = 32 g/mol

$$n = \frac{32}{16} = 2$$

So, M.F = $(NH_2)_2$ = N_2H_4

Q3 **A gaseous compound of nitrogen and oxygen was found to contain 7:16 by mass N and O respectively. Calculate molecular formula if vapour density is 46.**

A3 Vapour density = 46

Molar mass M_w = 46 g / mol

Gaseous compound of nitrogen and oxygen contains 7:16 by mass N and O.

Let mass of nitrogen = 7 x g

Mass of oxygen = 16 x g

Moles of nitrogen $= \frac{7x}{14} = \frac{x}{2}$

Moles of oxygen $= \frac{16x}{16} = x$

Simplest ratio = 1:2

E.F = NO_2

$$n = \frac{92}{46} = 2 \qquad \left[\therefore n = \frac{\text{Molecular mass}}{\text{Empirical formula mass}}\right]$$

$\Rightarrow$ M.F. = N_2O_4

Solution and Concentration Terms

SOLUTION

Solution is a homogeneous mixture of two or more non-reactive substances.
e.g. sugar syrup, air etc.

Every solution can be divided into two components only as solute and solvent.
i.e. Solute + solvent = solution
Solute < solvent

A solution can have one or more solutes but number of solvent is always one.

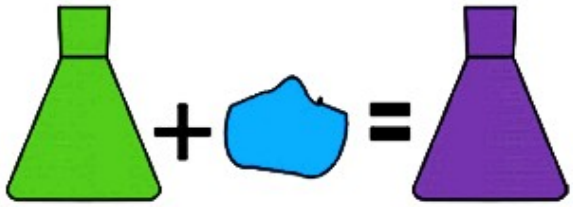
Binary solution	**= 1 solute + 1 solvent**
Ternary solution	**= 2 solute + 1 solvent**
Quaternary solution	**= 3 solute + 1 solvent**

But $W_{solution} = W_{solute} + W_{solvent}$

$V_{solution} = V_{solute} + V_{solute}$
$\neq V_{Solute} + V_{solvent}$
} both case will be possible in case of volume

In dilute solutions,
Solute <<<Solvent
i.e. $W_{Solution} = W_{Solvent}$
and $V_{solution} = V_{solvent}$

In aqueous solution, water is solvent. The composition of a solution can be expressed by various concentration terms which are as following:



1. MOLE FRACTION (X)

$$\text{Mole fraction of solute} = \frac{\text{mole of solute}}{\text{mole of solute} + \text{mole of solvent}}$$

$$\text{i.e.} \left[X_{\text{solute}} = \frac{n_{\text{solute}}}{n_{\text{solute}} + n_{\text{solvent}}} \right] < 1$$

$$\text{Mole fraction of solvent} = \frac{\text{mole of solvent}}{\text{Total moles}}$$

$$\text{i.e.} \left[X_{\text{solvent}} = \frac{n_{\text{solvent}}}{n_{\text{total}}} \right] < 1$$

$\boxed{X_{\text{solute}} + X_{\text{solvent}} = 1}$ for binary solution

e.g. If $X_{\text{solute}} = 0.1 = \frac{1}{10}$

Then moles of solute = 1
and moles of solvent = 10-1 = 9 moles

2. % (W/W) OR PERCENT WEIGHT BY WEIGHT

Mass of solute (gm) present in 100 gm of solution

$$\%w/w = \frac{W_{\text{solute(gm)}}}{W_{\text{solution(gm)}}} \times 100$$

DERIVATION

⇒ Suppose 'W' gram solution contains 'x' g solute then 100 g solution

$$\text{Contains solute} = \frac{x}{w} \times 100$$

⇒ For x% W/W solution
x gm solute present in 100 g solution then solvent =(100-x)g.

3. % (V/V)

It is defined as the volume of solute (ml) present in 100 ml of solution.

$$\%(V/V) = \frac{\text{volume of solute(ml)}}{\text{volume of solution(ml)}} \times 100$$

For x %(V/V) solution mean x ml solute in 100 ml of solution

4. %(W/V)

It is defined as the mass of solute (gm) in 100 ml of solution.

$$\%(W/V) = \frac{\text{mass of solute(gm)}}{\text{volume of solution(ml)}} \times 100$$

⇒ For x %(w/v) solution means x gram of solute in 100 ml of solution.

5. Parts per million (ppm)

It is defined as mass of solute (gm) in 10^6 gm of solution.

$$ppm = \frac{W_{solute(gm)}}{W_{solution(gm)}} \times 10^6$$

⇒ x ppm solution means 'x' gm solute in 10^6 gm solution,

solvent = $(10^6 - x)$ gm.

6. MOLARITY (M)



It is defined as moles of solute present in 1 litre of solution.

$$\text{Molarity} = \left(\frac{n_{solute}}{V_{solution}\text{ in L}}\right)\frac{\text{mol}}{\text{L}} \text{ or molar}$$

$$\text{Molarity} \times V_{solution} = n_{solute}$$

⇒ If volume of solution is in litres, then we will get moles.

⇒ If volume of solution is in mL, then we will get millimoles.

[X] = y [M]

i.e. Molarity of solution of X=y moles / L.

⇒ y mole of solute in 1 litre of solution.

I deal with undissociated species. I tell the concentration of the original solute.

Formality

7. MOLALITY (M)

If is defined as the moles of solute present in 1 kg of solvent

$$\text{Molality} = \frac{\text{Moles of solute}}{\text{Mass of solvent(kg)}}$$

mol kg^{-1} or molal

or $m = \frac{n_{solute}}{w_{solvent}}$ molal

⇒ for 'x' molal solution

i.e. x moles of solute in 1kg of solvent

I deal with compounds that ionize in solution. After ionization, the parent compound no more exists. So I give the concentration of the ions.

Formality

ILLUSTRATIONS:

Q1 **20 g of NaOH is present in its 100 ml aqueous solution. If density of solution is 1.1 gm/ml then calculate.**

i) % w/v

ii) % w/w

iii) Mole fraction

A1 Given, mass of NaOH = 20 g

Volume of solution = 100 ml

density = 1.1 g / ml

Mass of solution = v×d =100×1.1 g/ml = 110 gm

$$\text{moles of NaOH} = \frac{20}{40} = 0.5 \text{ mole}$$

Mass of solvent = 110-20 = 90 gm

$$\text{Moles of solvent} = \frac{90}{18} = 5 \text{mole}$$

i) $$\%/w/v = \frac{w_{solute}}{vol_{sol}} = \times 100 = \frac{20}{100} \times 100 = 20\%$$

ii) $$\%/w/w = \frac{w_{solute}}{w_{solution}} \times 100 = \frac{20}{110} \times 100 = \frac{200}{11}\%$$

iii) $$X_{NaOH} = \frac{n_{solute}}{n_{solute} + n_{solvent}} = \frac{0.5}{5 + 0.5} = \frac{0.5}{5.5} = \frac{1}{11}$$

Q2 **6.02×10^{20} molecules of urea are present in 100 ml solution. The concentration of urea solution is**

(A) 0.1 M **(B)0.01 M**

(C) 0.02 M **(D)0.001 M**

A2 **(B)**

No. of molecules of urea = 6.02×10^{20}

Volume of solution = 100 ml

$$\text{Moles of urea} = \frac{6.02\times10^{20}}{6.02\times10^{23}} = 10^{-3} \text{ mole}$$

$$[\text{Urea}] = \frac{\text{moles of urea}}{\text{vol. of solution}} = \frac{10^{-3}}{\frac{100}{1000}} = 10^{-2} \text{ M}$$

Q3 **Dissolving 120 g of urea (mol. wt. 60) in 1000 g of water gave a solution of density 1.15 g/ml. The molarity of the solution is:**

(A) 1.78 M **(B) 2.00 M**

(C) 2.05 M **(D) 2.22 M**

A3 **(C)**

Mass of urea = 120 g

Molecular wt. of urea = 60

$$\text{Moles of urea} = \frac{120}{60} = 2 \text{ moles}$$

Mass of solvent (H_2O) = 1000 g

Mass of solution = $m_{solute} + m_{solvent} = 120+1000 = 1120$ gm

$$\text{Volume of solution} = \frac{\text{mass}}{\text{density}} = \frac{1120}{1.15}\text{ml} \quad \left[\because d = \frac{m}{v} \text{ and } v = \frac{m}{d}\right]$$

$$\text{Using, Molarity} = \frac{n_{solute}}{v_{solution}}$$

$$\text{Molarity of urea, [Urea]} = \frac{2\times1.15\times1000}{1120} = \frac{1.15\times100}{56} = \frac{115\times100}{100\times56} = 2.05$$

Q4 **A 20% (w/v) solution has density 1.25 g/ml. The %(w/w) for the solution will be**

(A) 16 **(B) 25**

(C) 40 **(D) 12**

A4 **(A)**

Given, 20% (w/v) solution means 20 gm solute in 100 ml solution

density = 1.25 g/ml

$$\text{Mass of solution} = 100\times1.25 = 125 \text{ g} \quad \left[\because d = \frac{m}{v}, \quad m = v\times d\right]$$

$$\%\ (w/w) = \frac{w_{solute}}{w_{solution}}\times100 = \frac{20}{125}\times100 = 16\%$$

Q5 **A 5.2 molal solution of methyl alcohol, CH_3OH, is supplied. What is the mole fraction of methyl alcohol in the solution?**

(A) 0.100 **(B) 0.190**

(C) 0.086 **(D) 0.050**

A5 **(C)**

Given, 5.2 mole CH_3OH in 1 kg H_2O

$$\text{Moles of } H_2O = \frac{1000}{18} = 55.5$$

$$\text{Mole fraction of } CH_3OH = \frac{5.2}{5.2 + 55.5} \qquad \left[\because x = \frac{n_{solute}}{n_{solute} + n_{solvent}}\right]$$

$$= \frac{5.2}{60.7} = \frac{52}{60.7} = 0.087$$

Q6 **The density (in g mL^{-1}) of a 3.60 M sulphuric acid solution that is 29% H_2SO_4 (molar mass = 98 g mol^{-1}) by mass will be**

(A) 1.64 **(B) 1.88**

(C) 1.22 **(D) 1.45**

A6 **(C)**

Let density = x g/ml

3.6 mole H_2SO_4 in 1 L solution

weight of solution = 1000×'x' =1000 x gm $\qquad \left[\therefore d = \frac{m}{v} \;,\; m = d \times v\right]$

$$\%(w/w) = \frac{w_{solute}}{w_{solution}}$$

$$29 = \frac{3.6 \times 98}{1000\ x} \times 100$$

$$\Rightarrow \quad x = \frac{3.6 \times 98}{10 \times 29} = 1.22$$

$$x = 1.22$$

Q7 **A compound H_2X with molar weight of 80 g is dissolved in a solvent having density of 0.4 g ml^{-1}. Assuming no change in volume upon dissolution. The molality of a 3.2 molar volume is**

A7 Given density = 0.4 g ml^{-1}
and 3.2 mole in 1 litre solution
No change in volume i.e. $v_{solution} = v_{solvent}$

$$1 \text{ litre} = v_{solvent}$$

Mass of solvent = volume × density = 1000×0.4 = 400 gm

$$\text{Molality} = \frac{3.2}{400/1000} = 8 \text{ mol kg}^{-1}$$

Q8 **The mole fraction of a solute in a solution is 0.1. At 298 K molarity of this solution is the same as its molality. Density of this solution at 298 K is 2 g cm^{-3}. The ratio of the molecular weights of the solute and solvent is?**

A8 Let number of moles of solute = 1 mole,
mass of solute = 1 × 'X' g
X = molecular weight of solute
Y = molecular of solvent

Total number of moles = 10 mole
moles of solvent = 10-1 = 9 mole
mass of solvent = 9×y gram
mass of solution = mass of solute + mass of solution
= (X+9Y)g

Volume of solution $= \frac{X+9Y}{2} ml \quad \left[\because v = \frac{m}{d}\right]$

According to the question,

Molarity = Molality

$$\frac{1\times1000}{\frac{(X+9Y)}{2}} = \frac{1\times1000}{9Y}$$

$$18y = X + 9Y$$

$$(18-9)Y = X \Rightarrow 9Y = X$$

$$\Rightarrow \quad \frac{X}{Y} = \frac{9}{1}$$

Q9 **The mole fraction of urea in an aqueous urea solution is 0.05. If the density of the solution is 1.2 g cm^{-3}, the molarity of urea solution is**

A9 Given, mass of water = 900 g

Mole fraction of urea = 0.05 = $\frac{1}{20}$

let urea in solution = 1 mole

Then mole of H_2O = 20-19 = 19 mole

Mass of urea = 60×1 = 60 gm

Mass of H_2O = 19×18 = 342 gm

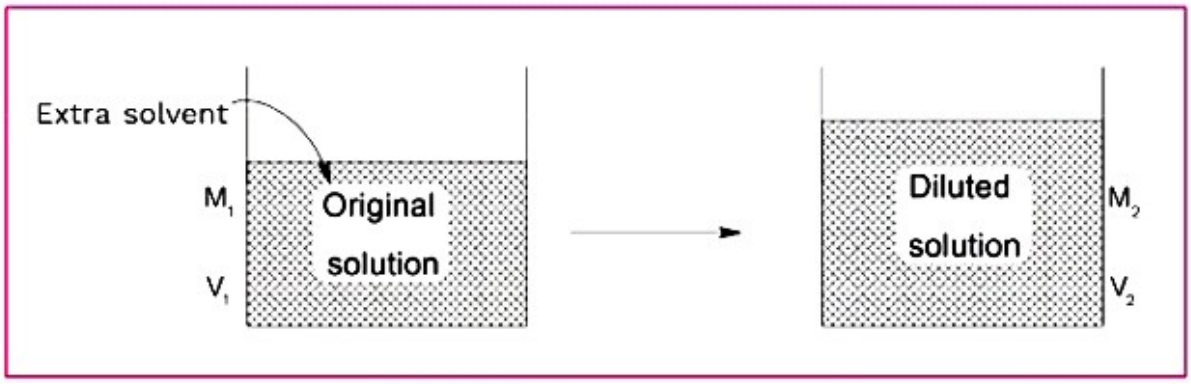
Mass of solution = (342+60)g = 402 g

Volume of solution = $\frac{402}{1.2}$mL

$$\text{Molarity} = \frac{n_{solute}}{v_{solution}} = \frac{1\times1000}{\frac{402}{1.2}} = \frac{1.2\times1000}{402} = \frac{1200}{402} = \frac{600}{201} = \frac{200}{67} = 2.98$$

DILUTION OF SOLUTION

Addition of extra solvent into a solution



Upon dilution, amounts of solvent and solution increase but **amount of solute remains constant** and due to which values of all concentration are changed (↓)

For dilution,

$$\boxed{M_1V_1 = M_2V_2}$$

Where M_1 is molarity before dilution

V_1 is volume of solution before dilution

M_2 is molarity after dilution

And V_2 is volume of solution after dilution.

ILLUSTRATIONS:

Q1 **200 ml of 1M aqueous NaCl solution is diluted by 800 ml. Calculate molarity of diluted solution.**

A1 Given , $V_1 = 200$ $V_2 = 800 + 200 = 1000$ ml

$M_1 = 1M$ $M_2 = ?$

Using, $M_1V_1 = M_2V_2$

$M_1 \times 200 = M_2 \times 1000$

$$\Rightarrow \quad M_2 = \frac{1 \times 200}{1000} = \frac{2}{10} \qquad M_2 = 0.2 \text{ M}$$

Q2 **How much volume of water must be added to 200 ml of 0.65 M HCl to dilute the solution to 0.20 M?**

A2

Given , $V_1 = 200$ ml $M_2 = 0.20$ M

$M_1 = 0.65$ M $V_2 = ?$

Using, $M_1V_1 = M_2V_2$

$0.65 \times 200 = 0.20 \times V_2$

$$\Rightarrow \quad V_2 = \frac{0.65 \times 200}{0.20} = \frac{65 \times 200}{20}$$

$V_2 = 650$ ml (Final solution)

Added volume $= V_2 - V_1$

$= 650 - 200 = 450$ ml

Q3 **Determine the volume of dilute nitric acid (19% w/v) that can be prepared by diluting 50 ml of conc. HNO_3 (76% w/v) with water**

A3 Given, Original solution → 50 ml of conc.HNO_3 (76%w/v) with water
Final solution → diluted HNO_3(19%w/v)

$$M_1 = \frac{\frac{76}{63}}{\frac{100}{1000}} \qquad M_2 = \frac{\frac{19}{63}}{\frac{100}{1000}}$$

Using $M_1V_2 = M_2V_2$

$$\frac{\frac{76}{63}}{\frac{100}{1000}} \times 50 = \frac{\frac{19}{63}}{\frac{100}{1000}} \times V_2$$

$$\frac{76\times1000}{63\times100}\times50 = \frac{19\times1000}{63\times100}\times v_2 \Rightarrow V_2 = \frac{76\times10\times50\times63}{63\times19\times10} = \frac{3800}{19} = 200 \text{ ml} \quad V_2 = 200 \text{ ml}$$

MIXING OF SOLUTIONS (NON-REACTING)

e.g. acid + acid
base + base
Salt + salt (one ion common)

Upon mixing, **volume of solution increases hence concentration of all solutes decreases** upon mixing.

$$\left[X_{Final} = \frac{n_x}{V_{final}}\right]$$

Where,

If ideal mixing then $[V_{final} = v_1 + v_2 + v_3 + ...]$

by default

If non- ideal mixing then $[V_{final} \neq v_1 + v_2 + ...]$

But $[W_f = w_1 + w_2 + ...]$

$V_fd_f = V_1d_1 + V_2d_2 + ...$

$$\Rightarrow v_f = \left[\frac{v_1d_1 + v_2d_2 + ...}{d_f}\right]$$

If d_1, d_2 ..... as well as d_f are given in question then only consider non-ideal mixing



ILLUSTRATIONS:

Q1 **The molarity of a solution obtained by mixing 750 ml of 0.5 M HCl with 250 ml of 2 M HCl will be**

(A) 1.00 M **(B) 1.75 M**

(C) 0.975 M **(D) 0.875 M**

A1 Given 750 ml of 0.5 M HCl mixed with 250 ml of 2M HCl

$$\text{Molarity} = \frac{n_{HCl}}{V_f} = \frac{(0.5\times 750)+(2\times 250)}{750+250}\frac{\text{m mole}}{\text{mL}}$$

$$= 0.875\ \text{M}$$

Q2 **100 ml, 0.1 M HCl(aq) and 200 ml, 0.1 M H_2SO_4 solution are mixed together, then what is the final concentration of hydrogen ions in the mixture?**

A2

$$[HCl] = \frac{n_{HCl}}{V_f} = \frac{0.1\times 100}{300} = \frac{1}{30}\text{M}$$

$$[H_2SO_4] = \frac{n_{H_2SO_4}}{V_f} = \frac{(0.1\times 200)}{300} = \frac{1}{15}\text{M}$$

$$[H^+] = \frac{n_{H^+}}{V_f} = \frac{(0.1\times 100)\times 1+(0.1\times 200)\times 2}{300} = \frac{1}{6}\text{M}$$

Q3 **Determine [OH^-] in final solution prepared by mixing**

100 ml, 0.2 M NaOH(aq)	+	100 ml, (0.1M) $Ba(OH)_2$	+	300 ml $H_2O(l)$

A3

$$[OH^-] = \frac{(0.2\times 100)+(0.1+100)\times 2}{100+100+300} = \frac{2}{25}\text{M}$$

Water has low dissociation w.r.t alkalis [(NaOH and $Ba(OH)_2$)]

Q4 **If 2 M 200 ml HCl, 2 M 100 ml $CaCl_2$ and 5M 200 ml $AlCl_3$ are mixed then final concentration of Cl^- will be:**

(A) 2.5 M **(B) 3 M**

(C) 3.5 M **(D) 7.6 M**

A4 **(D)**

Given, 2M 200ml HCl, 2M 100 ml $CaCl_2$ and 5M 200 ml $AlCl_3$

$$[Cl^-] = \frac{(2\times200)1+(2\times100)2+(5\times200)3}{500} = \frac{400+400+3000}{500} = \frac{3800}{500} = \frac{38}{5} = 7.6\,M$$

STOICHIOMETRY

All amount related calculations for a chemical reaction are collectively known as stoichiometry.

$$\text{Reaction} \longrightarrow \text{Products}$$

It always requires balanced equation of the reaction.

In balanced equation, the number associated with all the substances is called **stoichiometric coefficient** of those substances.

e.g.

$$H_2 \longrightarrow 2H$$
$$N_2 + 3H_2 \longrightarrow 2NH_3$$

In above reaction, stiochiometric coefficient of H_2 and NH_3 are 2 and 2 respectively.

The value of stoichiometric coefficients of substance may be changed but the ratio among them is constant.

e.g
$$\left.\begin{aligned} &H_2 \longrightarrow 2H \\ &\tfrac{1}{2}H_2 \longrightarrow H \\ &2H_2 \longrightarrow 4H \\ &\tfrac{3}{2}H_2 \longrightarrow 3H \end{aligned}\right\} H_2 : H = 1:2 \text{ (constant)}$$

This ratio of stoichiometric coefficients gives information about the molar ratio in which reactants are reacted and products are formed.

For $H_2 \longrightarrow 2H$

If 1 mole $\longrightarrow$ 2 mole

4 mole $\longrightarrow$ 8 mole

5 mole $\longrightarrow$ 10 mole

In order to solve questions of stoichiometry

Given → Moles $\xrightarrow[\text{(Balanced reaction)}]{\text{Apply stoichiometric ratio}}$ Required → Moles

ILLUSTRATIONS:

Q1 **How many gm of HCl is needed for complete reaction with 43.5 gm MnO_2? (At. wt. of Mn= 55)**

$4HCl + MnO_2 \longrightarrow MnCl_2 + 2H_2O + Cl_2$

A1 Given mass of MnO_2= 43.5 gm

Moles of $MnO_2 = \frac{43.5}{87} = 0.5$ mole

For 1 mole of MnO_2 = 4 mole of HCl

0.5 mole of $MnO_2 = \frac{4}{1} \times 0.5$ mole of HCl = 2 mole

Molar mass of HCl = 2×36.5 = 73 gm

Or

$4HCl + MnO_2 \longrightarrow MnCl_2 + H_2O + Cl_2$

MnO_2: 43.5 gm ↓ 0.5 mole

Apply ratio 1:4

$= 4 \times 0.5$ mole

$= 2$ mole

$= 2 \times 36.5$ gm

$= 73$ gm

Q2 **How much volume of nitrogen dioxide (at STP conditions) is required with excess of water to produce 12.6 gm HNO_3?**
$3NO_2(g) + H_2O(l) \longrightarrow 2HNO_3(aq) + NO(g)$

A2

$$3NO_2(g) + H_2O(l) \longrightarrow 2HNO_3(aq) + NO(g)$$

$$12.6\,gm$$

$$\downarrow$$

$$\frac{12.6}{63}\,gm\,(moles\ of\ HNO_3)$$

$$0.3\ mole \xleftarrow{\text{Apply ratio 3:2}} = 0.2\,mole$$

Volume at STP = 0.3×22.4 = 6.72 L

Q3 **25 ml of solution of barium hydroxide requires 35 ml of 0.1M solution of hydrochloric acid. The molarity of barium hydroxide is:**

A3 $Ba(OH)_2(aq) + 2HCl(aq) \longrightarrow BaCl_2(aq) + 2H_2O(l)$

Base acid

Given, Volume of 0.1 M solution of HCl = 35 ml

Moles of HCl = 35×0.1 = 3.5 m mole

Applying ratio (1:2)

For 2 mole HCl $\longrightarrow$ 1 mole $Ba(OH)_2$

For 1 mole HCl $\longrightarrow$ 1/2 m moles of $Ba(OH)_2$

For 3.5 m mole HCl $\longrightarrow \frac{3.5}{2}$ m moles of $Ba(OH)_2$

$$M_{Ba(OH)_2} = \frac{n}{v} = \frac{\frac{3.5}{2}\,mmole}{25\,ml} = 0.07\ mol^{-1} = 0.07\,M$$

Concept of Limiting Reagent

Applicable in reactions having **two or more reactants.**
In such reactions, the reactant which gets finished very first, is called limiting reagent (L.R).
Reactants other than L.R are called **excess reagent.**
When L.R. gets finished, the reaction is stopped so now all calculations must be done w.r.t limiting reagent.
To determine limiting reagent, calculate value of following ratio for all reactants separately:

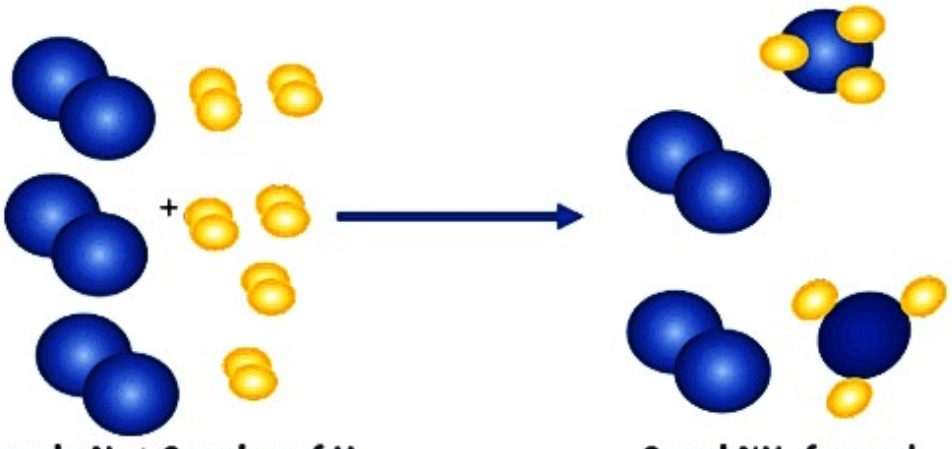
$$\left(\frac{\text{Initial mole}}{\text{stoichiometric coefficient}}\right)$$

THE REACTANT HAVING LEAST VALUE OF THIS RATIO WILL BE L.R.
Note: If ratio for all reactants are same then there will be no L.R means all reactants are completely finished simultaneously.

For the reaction $N_2 + 3H_2 \longrightarrow 2NH_3$

	Reactants	**Products**
Before reaction	**3 mole N_2 + 3 moles of H_2**	**0 mol NH_3**
After reaction	**3–1 mole N_2 is left**	**2 mol NH_3 formed**

H_2 acts as the limiting reagent and 2 mole of N_2 is left behind (excess reagent)



3 mole N_2 + 3 moles of H_2 **2 mol NH_3 formed**

ILLUSTRATIONS:

Q1 **For reaction: $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$, If 0.64 kg of each reactant are taken then**

(i) Calculate mass of H_2O formed

(ii) Calculate mass of CH_4 left after reaction

A1 $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$

Given, Mass of CH_4 = 0.64 kg = 640 g

Moles of CH_4 = 640 g /16gmol^{-1} = 40 mole

Moles of O_2 = $\frac{640}{32}$ = 20 mole

$\therefore$ O_2 is a limiting reagent

Applying ratio (O_2: H_2O) = (2:2)

For 20 mole of O_2 = 20 mole H_2O = 20×18 = 360g

(ii) Calculate remaining mass of CH_4

For 20 mole of O_2 = 10 mole of CH_4

Remaining CH_4 = Initial – reacted

= 40mole – 10 mole

= 30mole

= 480gm

Q2 **4 moles of N_2 and 14 moles of H_2 are allowed to convert into NH_3. The total number of moles of gases present after the reaction will be**

(A) 8 **(B) 2**

(C) 6 **(D) 10**

A2 **(D)**

$N_2 + 3H_2 \longrightarrow 2NH_3$

Given, 4 moles of N_2 and 14 moles of H_2

Applying ratio (N_2:H_2) = (1:3)

For 4 mole N_2= 12 mole of H_2

In this reaction, N_2 is limiting reagent and H_2 is excess reagent.

Remaining H_2 = Initial – reacted = (14–12) = 2 mole

Applying ratio (N_2: NH_3) = (1:2)

For 4 mole of N_2=8 mole of NH_3 (formed)

$\therefore$ Total number of moles of gases present after the reaction = 2+8 = 10 mole

Q3 **Calculate the vapour density of final mixture in previous question.**

A3 In final mixture, 2 moles of H_2 present and 8 moles NH_3 present

$$M_{mix} = \frac{w_{mix}}{n_{mix}} = \frac{2(2) + 8(17)}{2 + 8} = \frac{140}{10} = 14 \text{ g / mol}$$

$$(\text{Vapour density})_{mix} = \frac{\text{molar mass}}{2} = \frac{14}{7} = 7$$

Q4 **50ml, 0.2M HCl(aq) is mixed with 50 ml, 0.1 M NaOH(aq). Find the nature of final solution.**

A4 $\underset{\text{acidic}}{HCl} + \underset{\text{basic}}{NaOH} \longrightarrow \underbrace{NaCl + H_2O}_{\text{Neutral}}$ Given: 50×0.2 = 10 m mole HCl

50×0.1 = 5 m mole NaOH

Here, NaOH is limiting reagent.

So, final solution contains only HCl and NaCl and H_2O (i.e. acidic, neutral, neutral).

So, final solution is acidic in nature.

Q5 **Equal volumes of 0.1M H_2SO_4(aq) and 0.1M KOH(aq) solutions are mixed together. The final molarity of H^+ or OH^- will be**

(A) 0.1 M **(B) 0.05 M**

(C) 0.2 M **(D) 0.25 M**

A5 **(B)**

$H_2SO_4 + 2KOH \longrightarrow K_2SO_4 + 2H_2O$

Let V L of 0.1M H_2SO_4 is mixed with V L of 0.1 M KOH

Moles of H_2SO_4 = 0.1 V mole

Moles of KOH = 0.1 V mole

$$\frac{0.1V}{1}_{(H_2SO_4)} > \frac{0.1V}{2}_{(KOH)} \qquad \left[\because \frac{\text{Inital mole}}{\text{Stoichiometric coefficien}}\right]$$

$\therefore$ KOH is a limiting reagent

Applying ratio (H_2SO_4: KOH) = 1:2

Remaining H_2SO_4 = Initial − Reacted = $0.1V - \frac{0.1V}{2} = \left(\frac{0.1V}{2}\right)$m mole

$$\left[H^+\right] = \frac{n_{H^+}}{V_f} = \frac{\left(\frac{0.1\ V}{2}\right)\times 2}{V+V} = 0.05M = \quad \text{\% yield of reaction}$$

$$\boxed{\text{\% Yield} = \frac{\text{observed amount of product}}{\text{theoretical amount of product}} \times 100}$$

Also,

$$\boxed{\text{Observed amount of product} = \frac{\text{\% yield}}{100} \times \text{theoretical amount}}$$

ILLUSTRATIONS:

Q1 **Cyclohexanol is dehydrated to cyclohexene on heating with concentrated H_2SO_4. If the yield of this reaction is 75% how much cyclohexene will be obtained from 100 gm of cyclohexanol?**

$C_6H_{12}O \xrightarrow{\text{conc.}H_2SO_4} C_6H_{10} + H_2O$

A1

$C_6H_{12}O \xrightarrow{\text{conc.}H_2SO_4} C_6H_{10} + H_2O$

Mass of $C_6H_{12}O$ = 100 gm

Moles of $C_6H_{12}O$ = 1 mole (theoretical)

Applying ratio ($C_6H_{12}O$: C_6H_{10}) = (1:1)

Moles of C_6H_{10}= 1 mole

Observed amount of cyclohexene = $\frac{75}{100} \times 1 = \frac{3}{4}$ mole

Mass of cyclohexene = 61.5 gm

Q2 **When 21 gm of Li reacts with 280 gm of N_2 then 0.35 gm of Li_3N is formed. What is percentage yield of the reaction?**

$6Li(s) + N_2(g) \longrightarrow 2Li_3N(s)$

A2

Observed amount of Li_3N = 0.35 gm (Given)

$6Li(s) + N_2(g) \longrightarrow 2Li_3N(s)$

Mole of Li= $\frac{21\,gm}{7\,gm\ mol^{-1}} = 3\,mole$

Mole of N_2 = $\frac{280\ gm}{28\,gm\,mol^{-1}} = 10\,mole$

Applying ratio (Li: Li_3N) = (6:2), we get

For 3 mole Li = 1 mole Li_3N

Mass of Li_3N = mole × molar mass = 1×35 = 35 g

% yield = $\frac{0.35}{35} \times 100 = 1\%$

PROBLEMS INVOLVING MIXTURE

Q1 **20 gm of a mixture of NaCl and NaOH exactly requires 7.3 gm HCl for complete reaction. The mass percent of NaCl in the original mixture is:**

(A) 40% **(B) 60%**

(C) 50% **(D) 80%**

A1

$HCl + NaOH \longrightarrow NaCl + H_2O$

Mass of HCl = 7.3 gm

Moles of HCl = $\frac{7.3\,g}{36.5\,g\,mol^{-1}} = 0.2$ mole

Applying (1:1) = (HCl: NaCl), we get

For 0.2 mole HCl = 0.2 mole NaOH

Let mass of NaOH = x gm

Moles of NaOH = 0.2 = $\frac{\text{mass}}{\text{molar mass}} \Rightarrow \quad 0.2 = \frac{x}{40} = 40 \times 0.2 = 8$ gm

Mass of NaCl = 20 – 8 = 12 gm

Mass of percent of NaCl in mixture = $\frac{12}{20} \times 100 = 60\%$

Q2 **39 gm of an alloy of Al and Mg when heated with excess of HCl forms magnesium chloride, aluminium chloride and hydrogen. The evolved hydrogen collected at 0°C has a volume of 44.8 l at 1 atm. Calculate the composition of the Al by moles.**

A2

Let moles of aluminium = x mole &Moles of magnesium = y mole

Mass of alloy = 39 gm

$\Rightarrow$ 39 gm = mass of Al + mass of Mg

$39 = x(27) + y(24)$

$13 = 9x + 8y$ ... (1)

$Al + 3HCl \longrightarrow AlCl_3 + \frac{3}{2}H_2$ ...(i)

$Mg + 2HCl \longrightarrow MgCl_2 + H_2$ ...(ii)

From reaction (i),

For x mole Al = $\frac{3}{2}$ x mole H_2

From reaction (ii),

For y mole Mg = y mole H_2

Total hydrogen (H_2) = $\left(\frac{3x}{2} + y\right)$mole $\quad \frac{44.8}{22.4}\text{mole} = \frac{3x}{2} + y$

$3x + 2y = 4$ ... (2)

$9x + 8y = 13$ ... (1)

Multiplying equation (2) by 3, we get

$9x + 6y = 12$ ... (3)

Subtracting equation (1) by (3)

$9x + 8y = 13$

$9x + 6y = 12$

$- \quad - \quad -$

$2y = 1 \quad y = \frac{1}{2} = 0.5$ = mole of Mg

wt of Mg = 12 $\Rightarrow$ wt of Al = 39–12 = 27g.

$n_{Al} = \frac{27}{27} = 1$ % moles of Al = $\frac{1}{1+0.5} \times 100 = 66.67\%$

Principle of Atoms Conservation (POAC)

In a chemical reaction, atoms of all elements remains conserved

	Reactants				⟶	products			
	C	H	N	O		C	H	N	O
Atoms	5	8	2	1		5	8	2	1

Even the balanced equation of reaction obeys POAC

$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$

C	H	O		C	H	O
1	4	4		1	4	4

Generally used when complete reactants is not given

Given $\xrightarrow{\text{atleast one element given conserved}}$ **required**

So, POAC on such element

Atoms of that element in given substance	=	atoms of that element in required substance

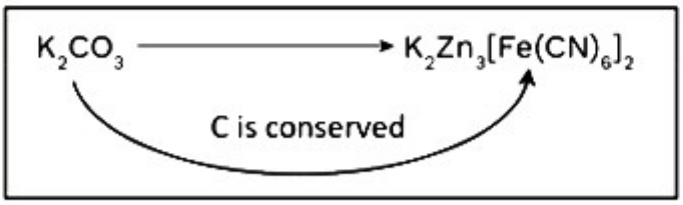
Principle of atoms conservation (POAC) is based on Laws of conservation of atom (Dalton's theory)

ILLUSTRATIONS:

Q1 **A 13.8 g sample of K_2CO_3 was treated in such a way that all of its carbon was captured in compound $K_2Zn_3[Fe(CN)_6]_2$. Compute the mass of this product in grams.**

A1 Given: mass of K_2CO_3= 13.8 g

K_2CO_3



POAC on carbon

$(n)_C = (n)_C$

[Atoms of that element in given substance = Atoms of that element in required substance]

$(n)_{K_2CO_3} \times 1 = (n)_{compound} \times 12$

$$\frac{13.8}{138} \times 1 = \frac{w}{698} \times 12$$

$$\frac{1}{10} \times \frac{698}{12} = w$$

$$\frac{698}{120} = w$$

$$w = 5.816$$

$$\approx 5.82$$

Q2 **In a determination of phosphorus an aqueous solution of NaH_2PO_4 is treated with a mixture of ammonium and magnesium ions to precipitate magnesium ammonium phosphate $Mg(NH_4)PO_4.6H_2O$. This is heated and decomposed to magnesium pyrophosphate $Mg_2P_2O_7$, which is weighed. A solution of NaH_2PO_4 yielded 3.33 gm of $Mg_2P_2O_7$. What weight of NaH_2PO_4 was present originally?**

A2 $NaH_2PO_4 \rightarrow\rightarrow\rightarrow\rightarrow Mg_2P_2O_7$

Mass of $Mg_2P_2O_7$= 3.33 gm (given)

POAC on P

$$(n)_P = (n)_P$$

$$(n)_P = (n)_P$$

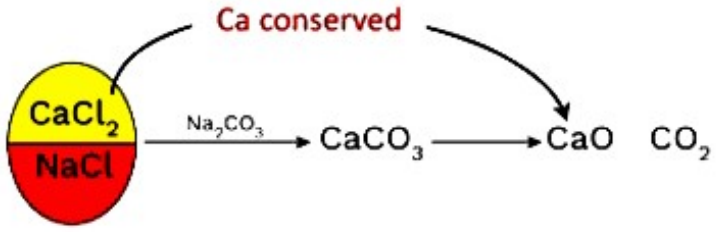
$$(n)_{NaH_2PO_4} \times 1 = (n)_{Mg_2P_2O_7} \times 2$$

$$\frac{w}{120} \times 1 = \frac{3.33}{222} \times 2$$

$$w = 3.6 \text{ gm}$$

Q3 **A 10gm sample of a mixture of calcium chloride and sodium chloride is treated with Na_2CO_3 to precipitate calcium as calcium carbonate. This $CaCO_3$ is heated to convert all the calcium to CaO and the final mass of CaO is 1.12 gm. Calculate the mass of NaCl in the original mixture.**

A3



Let mass of $CaCl_2$= w gm

POAC on Ca, $(n)_{Ca} = (n)_{Ca}$

$$\frac{w}{111} \times 1 = \frac{1.12}{56} \times 1$$

$$w = \frac{1.12}{56} \times 111$$

$$w = 2.22\,\text{gm}$$

Mass of $CaCl_2$= 2.22 gm

Mass of NaCl = 10−2.22 = 7.78 g

Q4 **Sulphur trioxide may be prepared by the following two reactions:**

$S_8 + 8O_2(g) \longrightarrow 8SO_2(g)$

$2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$

How many grams of SO_3 will be produced from 1 mole of S_8?

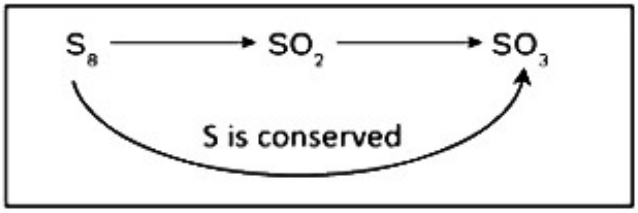
A4 Given 1 mole of S_8

POAC on S

$(n)_s = (n)_s$

$1 \times 8 = \frac{w}{80} \times 1$

w = 640 gm

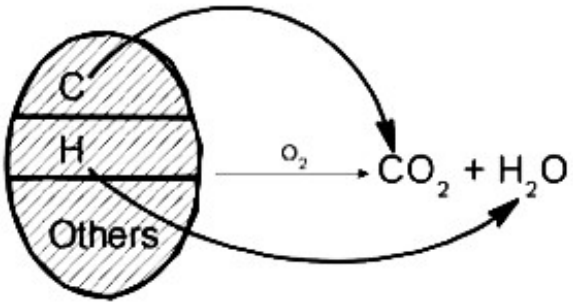


Q5 **On complete combustion, 0.1 g of an organic compound gave 0.176 g of carbon dioxide and 0.18g of water. Determine the percentage composition of carbon and hydrogen in the compound**

A5 Given mass of organic compound = 0.1g

Mass of CO_2 = 0.176 g

and mass of H_2O=0.18 g



POAC on carbon

$(n)_c = (n)_3$

$\frac{w_c}{12} \times 1 = \frac{0.176}{44} \times 1$

$w_c = \frac{0.176}{44} \times 12 = 0.048\,g$

POAC on H $(n)_H = (n)_H$

$\frac{w_H}{1} \times 1 = \frac{0.18}{18} \times 2 \times 1$

$= 0.020\,gm$

% composition of C = 48%

% composition of H = 20%

Q6 **By the reaction of carbon and oxygen, a mixture of CO and CO_2 is obtained. What is the mass ratio of the mixture obtained when 20 grams of O_2 reacts completely with 12 grams of carbon?**

A6 Given:

C and $O_2 \xrightarrow{\text{Completely}} CO$ & CO_2

Given mass of C = 12 grams and mass of O_2=20 grams

Moles of CO = x mole

Moles of CO_2 = y mole

POAC on C

$$(n)_c = (n)_c$$

$$1 \times 1 = x(1) + y(1)$$

$$x + y = 1 \qquad \ldots (1)$$

POAC on O

$$(n)_o = (n)_o$$

$$\frac{5}{8} \times 2 = x(1) + y(2) \qquad \Rightarrow x + 2y = \frac{5}{4} \quad \ldots(2)$$

Subtracting equation (2) by (1) we get

$$\begin{array}{lllll} x & + & 2y = & \frac{5}{4} \\ x & + & y = & 1 \\ - & & - & - \\ \hline & & y = \frac{1}{4} \end{array}$$

$x+y = 1$

Putting $y=\frac{1}{4}$, we get $\left[x = \frac{3}{4}\right]$

$$\frac{w_{CO}}{w_{CO_2}} = \frac{\frac{3}{4} \times 28}{1/4 \times 44} = \frac{3 \times 28 \times 4}{4 \times 44} = \frac{3 \times 7}{11} = \frac{21}{11}$$

Q7 **6 gm of silver salt of tribasic acid gives 4.32 gm silver on strong heating. The molar mass of acid is**

(A) 126 gm **(B) 129 gm**

(C) 123 gm **(D) 252 gm**

A7 **(B)**

Let acid H_3C (Molar mass = (3+x) gm/mol)

Silver salt = Ag_3X

Now, $Ag_3X \xrightarrow{x} Ag$

POAC on Ag $(n)_{Ag} = (n)_{Ag}$

$$\left(\frac{6}{324+x}\right)\times 3 = \frac{4.32}{108}\times 1$$

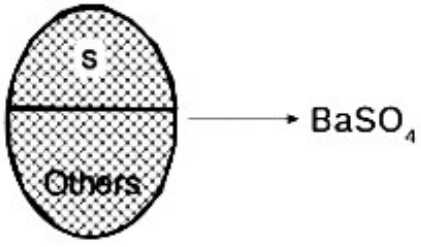
$x = 126$ g/mol

Molar mass of acid $= 3+x$

$= 3+126 = 129$ g / mol

Q8 **In sulphur estimation, 0.64 g of an organic compound gave 0.466 g of barium sulphate. What is the percentage of sulphur in the compound?**

A8 Let mass of sulphur = x gm



POAC on S $(n)_s = (n)_s$

$$\frac{x}{32}\times 1 = \frac{0.466}{233}\times 1$$

$$x = \frac{0.466}{233}\times 1\times 32$$

$X = 64\times10^{-3}$ gm

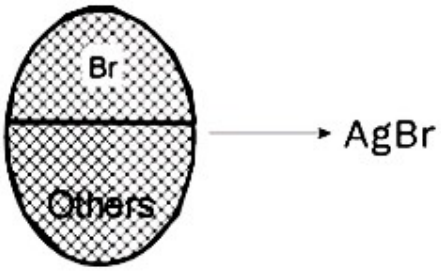
$$\% \text{ of S} = \frac{64\times10^{-3}}{0.64}\times 100 = 10\%$$

Q9 **In carius method of estimation of halogens, 250mg of an organic compound gave 141 mg of AgBr. The percentage of bromine in the compound is**

(A) 48 **(B) 60**

(C) 24 **(D) 36**

A9 Let mass of Br = x gram



POAC on Br $(n)_{Br} = (n)_{Br}$

$$\frac{x}{80}\times 1 = \frac{141}{188}\times 1$$

$$x = \frac{141}{188}\times 80$$

$$x = 3\times 20$$

x= 60

$$\% \text{ of Br} = \frac{x}{250}\times 100 = \frac{60}{250}\times 100 = 24\%$$

Q10 **During estimation of nitrogen present in organic compound by Kjeldahl's method, the amount evolved from 0.5g of compound requires 100 ml of 0.1M H_2SO_4(aq). Find out the percentage of nitrogen in the compound.**

A10 Given mass of compound = 0.5 g

Volume of 0.1M H_2SO_4 = 100 ml



$$\xrightarrow{} NH_3$$

$$2NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$$

Apply ratio (NH_3:H_2SO_4) = (1:2)

For 10 m mole of H_2SO_4 $\longrightarrow$ 20 m mole of NH_3

POAC on N $\quad (n)_N = (n)_N$

$$\frac{x}{14} \times 1 = 20 \times 10^{-3}$$

x=0.28 gm

$$\% \text{ of N} = \frac{0.28}{0.5} \times 100 = 56\%$$

Sequential Reactions

Simultaneously running multiple reactions in which one of the product acts as one of the reactant for next reaction.

i) $A \longrightarrow 2B + 3C$

If 10 mole 20 mole 30 mole is given

$B \longrightarrow 2C + D$

20mole 40mole 20mole

i.e. C=30 + 40 =70 mole

D = 20 Mole

OR

$A \longrightarrow 2(2C + D) + 3C$

$A \longrightarrow 7C + 2D$

10 mole 70 mole 2 mole

ILLUSTRATION:

Q1 **$KClO_4$ can be prepared by Cl_2 and KOH by a series of reactions as given below.**

$Cl_2 + 2KOH \longrightarrow KCl + KClO + H_2O$

$3KClO \longrightarrow 2KCl + KClO_3$

$4\ KClO_3 \longrightarrow 3KClO_4 + KCl$

i) Calculate mass of Cl_2 required to produce 1385 g $KClO_4$

ii) Calculate the total mass of KCl produced is 1st, 2nd and 3rd reaction

iii) Calculate the mass of $KClO_4$ produced by 142.9 gm Cl_2.

A1 i) (2840)gm

ii) (5217)gm

iii) (69.25)gm

Sol. i) $Cl_2 + 2KOH \longrightarrow KCl + KClO + H_2O$

x mole x mole

$3KClO \longrightarrow 2KCl + KClO_4$

X mole x/3 mole

$4KClO_3 \longrightarrow 3KClO_4 + KCl$

$\frac{x}{3}$ mole $\quad \frac{3}{4}\left(\frac{x}{3}\right)$

$= \frac{x}{4}$ mole

Mass of $KClO_4 = \frac{x}{4} \times 138.5 = 1385$

$x = \frac{1385 \times 4}{138.5}$

moles of $Cl_2 = 40$

Mass of $Cl_2 = 40 \times 71$ gm

$= 2840$ g

ii) $Cl_2 + 3KOH \longrightarrow KCl + KClO + H_2O$

40 mole 40 mole

$3KClO \longrightarrow 2\,KCl + KClO_3$

40 moles $\frac{80}{3}$ mole

$4KClO \longrightarrow 3KClO_4 + KCl$

$\frac{40}{3}$ mole $\quad \frac{1}{4} \times \frac{40}{3}$ mole

Total moles of KCl = $40 + \frac{80}{3} + \frac{10}{3} = \frac{120 + 80 + 10}{3} = \frac{210}{3}$

Mole of KCl = 70×74.5 gm

= 5215 gm

iii) Moles of $Cl_2 = \frac{142.9\,g}{70.99\,g/mol} = 2$ mole

$Cl_2 + 2KOH \longrightarrow KCl + KClO + H_2O$

2 mole 2 mole

$3\,KClO \longrightarrow 2KCl + KClO_3$

2 mole $\quad \frac{1}{3} \times 2 = \frac{2}{3}$ mole

$4\,KClO_3 \longrightarrow 3\,KClO_4 + KCl$

$\frac{2}{3}$ mole $\quad \frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$ mole = 0.5 mole

Moles of $KClO_4$ = 0.5 mole

Mass of $KClO_4$ = 0.5× 138.5 g = 69.25 g

Q2 **For following sequence of reaction**

$$4NH_3 + 5O_2 \xrightarrow{90\%} 4NO + 6H_2O$$
$$2NO + O_2 \xrightarrow{80\%} 2NO_2$$
$$3NO_2 + H_2O \xrightarrow{85\%} 2HNO_3 + NO$$

Starting with 3 moles of NH_3 what are moles of HNO_3 produced?

A2 $4NH_3 \xrightarrow{90\%} 4NO$

$$\text{Moles of } NH_3 = \frac{51gm}{17\ g\ mole^{-1}} = 3 mole$$

$$\Rightarrow \quad \text{for 3 mole of } NH_3 = \frac{90}{100} \times 3 \text{ mole of NO} \quad [\because NH_3 : NO = (1:1)]$$

$$= 2.7 \text{ mole of NO}$$

$$2NO + O_2 \xrightarrow{80\%} 2NO_2$$

$$\text{For 2.7 mole NO} = \frac{80}{100} \times 2.7 \text{mole of } NO_2 \quad [\because NO : NO_2 = (1:1)]$$

$$= 2.16 \text{ mol of } NO_2$$

$$3NO_2 + H_2O \xrightarrow{85\%} 2HNO_3 + NO$$

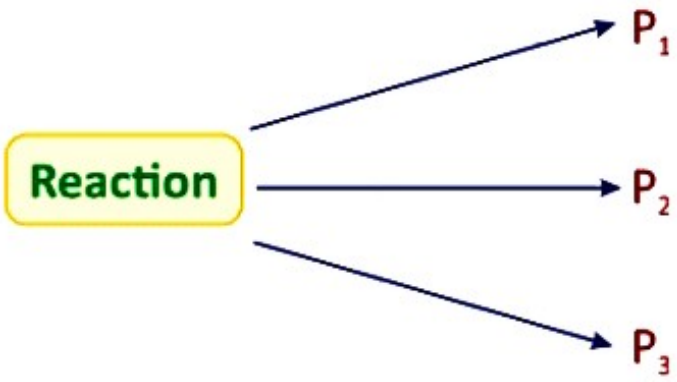
$$\text{For 2.16 mole of } NO_2 = \frac{2}{3} \times 2.16 \times \frac{85}{100} \text{ mole of } HNO_3 = 1.22 \text{ mole}$$

$$\text{Mass of } HNO_3 = 1.22 \times 63.01 \text{ g/mol}$$
$$= 76.87 \approx 77 \text{ g}$$

PARALLEL REACTION

The reactions in which a substance reacts or decompose in more than one way are called parallel reactions

FOR EXAMPLE



FOR EXAMPLE

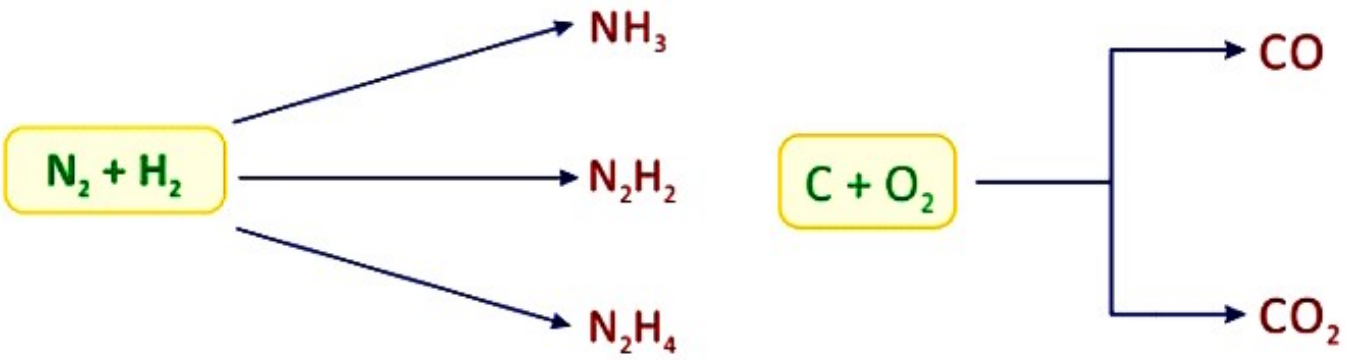


ILLUSTRATION:

Q3 **By the reaction of carbon and oxygen, CO or CO_2 or both are formed. Determine composition of final mixture if 96 g O_2 is allowed to react with 24 g carbon.**

$$2C + O_2 \longrightarrow 2CO$$
$$C + O_2 \longrightarrow CO_2$$

A3 Mole of C = $\frac{24g}{12g\ mol^{-1}} = 2mole$

Here carbon is L.R

Moles of $O_2 = \frac{96g}{32g/mole^{-1}} = 3mole$

Remaining O_2 = Initial – consumed

= 3 –1 = 2 mole

Now, 2 mole CO and 2 mole O_2formed

$2CO + O_2 \longrightarrow 2CO_2$

2 mole 2 mole 2

(L.R)

Remaining O_2= (2–1) = 1 mole

Now, 2 mole CO_2 and 1 mole O_2

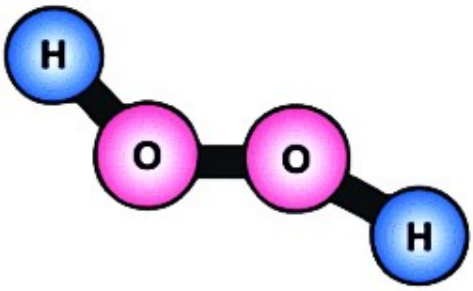
Volume Strength of H_2O_2

A special concentration term only used for H_2O_2 solution

At room temperature, H_2O_2 decomposes itself as given by following reaction

$H_2O_2 \longrightarrow H_2 + O_2$

Structure of H_2O_2



$$\text{Volume strength} = \frac{\text{volume of } O_2 \text{ at STP(L)}}{\text{vol. of } H_2O_2 \text{ solution taken (L)}}$$

It is a unit less quantity
It gives information about the molarity of H_2O_2 solution

DERIVATION:

Let $[H_2O_2]$ = x M and V_{sol} = V litre

$$H_2O_2 \longrightarrow H_2O + \frac{1}{2}O_2$$

(xV)mole $\left(\frac{xV}{2}\right)$mole

(New convention)

Now, volume of O_2 at STP $= \frac{xV}{2} \times 22.7L$

So, volume strength $= \frac{xV}{2} \times 22.7L$

$$V.S = \frac{\frac{xV}{2} \times 22.7L}{VL} = x \times 11.35$$

(Old convention)

V of O_2 at STP $= \frac{xV}{2} \times 22.4L$

$$V.S = \frac{\frac{xV}{2} \times 22.4L}{VL} \quad x = 11.2$$

i.e. Volume strength= 11.35× molarity
or
= 11.2× molarity

Representation

If V.S = 2 ⇒ 2V H_2O_2solution
Or
2 volume H_2O_2solution

ILLUSTRATION:

Q1 **The volume strength of 1M H_2O_2 is [JEE Mains 2019 Jan]**
(A) 22.4 **(B) 11.35**
(C) 16.8 **(D) 5.6**

A1 **(B)**
Volume strength of H_2O_2 = 11.35× molarity
= 11.35×1
= 11.35

Q2 **The strength of 11.2 volume solution of H_2O_2 is**
(A) 13.6% **(B) 3.4%**
(C) 1.7% **(D) 34%**

A2 **(B)**
Volume strength = 11.2
Volume strength = 11.2 × molarity (old convention)

$$\text{Molarity} = \frac{11.2}{11.2} = 1 \text{ mole/litre}$$

1 mole H_2O_2 in 1 L solution

$$\%\ (w/v) = \frac{1\times 34}{1000}\times 100 = 3.4\%$$

Q3 **500 ml of a H_2O_2 solution on complete decomposition produces 2 moles of H_2O. Calculate the volume strength of H_2O_2 solution?**

A3 Given, 2 moles of H_2O_2 produced means 1 mole O_2 produced

i.e. 2 mole $H_2O_2 \Rightarrow$ 1 mole O_2

$= 1 \times 22.7$ L

$$V.S = \frac{\frac{22.7}{500}\text{L}}{\frac{1}{1000}} = 45.4$$

EUDIOMETRY

It simply means analysis of gases components in a gaseous phase reaction.

An experimental produce used for gaseous reaction only. (homogeneous as well as heterogeneous) which take place at constant temperature and pressure.

Monitoring of these reactions is done in terms of volume of gases (mL or L)

∵ for gases,

$PV = nRT$

$V \propto n$ (∵ P and T constant)



∴ Stoichiometry ratio will give information of volume ratio for gases

i.e. A(g)	⟶	2B(g)
If 10 ml		20 ml
25		50 ml

Let V_1 is total volume before reaction

V_2 is total volume after reaction

If volume contraction takes place then $V = (V_1 - V_2)$

If volume expansion takes place then $V = (V_2 - V_1)$

If no volume change then $V_1 = V_2$

Note : In combustion of a gaseous substance,

$V_1 = (V)_{substance} + (V)_{required\ O_2}$

SPECIAL POINTS:

1. Until given, specifically consider room temperature and pressure

2. Consider volumes of solids and liquid to be zero

3. Consider air as a mixture of N_2 and O_2 only

   (Unit given: 20% O_2 + 80% N_2)

4. Until given, consider N_2 as non – reactive

5. There are some substances which absorb particular gases when gases are passed through them

SUBSTANCE	GAS ABSORBED
KOH(aq), NaOH(aq)	CO_2(g), HCl(g)
Alkaline, Pyragallol	O_2(g)
Turpentine oil	O_3(g)
Anhydrous $CaCl_2$	H_2O(g)
Water	NH_3(g), HCl(g)

ILLUSTRATION:

Q1 **50 ml of C_3H_8 is mixed with 300 ml of O_2 for complete combustion then find the final volume and volume contraction.**

A1 $C_3H_8(g) + 5O_2(g) \longrightarrow 3CO_2(g) + 4H_2O(l)$

50 ml 300 ml

(L.R)

Remaining O_2 = Initial – consumed

= 300 – 250 = 50 ml

Volume of CO_2 = 150 ml and volume H_2O is neglected

V_1=50 + 300 = 350 ml

V_2= 50 + 150 = 200 ml

i.e. volume decreases

$V = V_1 - V_2$ = 350 – 200 = 150 ml

Q2 **10 litre of N_2 and 40 litre of H_2 were reacted to give ammonia at constant T and P. Calculate change in volume due to reaction if 50% reaction takes place.**

A2

	$N_2(g)$	+	$3H_2(g)$	$\longrightarrow$	$2NH_3(g)$
Initially	10 L		40 L		
(L.R)					
At 50% reaction	5 L		(40–15) =25 L		10 L

Remaining = (10–5L) = 5L N_2

V_1 = 10 + 40 = 50 L

V_2 = 5L+25 L + 10L= 40 L

i.e. volume decreases

$V = V_1 - V_2$

= 50 – 40 = 10 L

Q3 **10 ml of alkane on complete combustion gives 30 ml of CO_2. Find molecular formula of alkane.**

A3 30 ml of CO_2

$$C_nH_{2n+2} + \left(n + \frac{n+1}{2}\right)O_2 \longrightarrow nCO_2 + (n+1)\,H_2O$$

alkane

$10\,ml \xrightarrow[1:n]{} 10n = 30$

$$n = \frac{30}{10}$$

$$= 3$$

The molecular formula the alkane is $C_nH_{2n+2} = C_3H_8$

Q4 **5 ml of hydrocarbon on complete combustion gave 15 ml of CO_2 and if for this 20 ml of O_2 was required. Find molecular formula of hydrocarbon.**

A4

$$C_xH_y(g) + \left(x + \frac{y}{4}\right) O_2(g) \longrightarrow xCO_2(g) + \frac{y}{2}H_2O(l)$$

$5\,ml \xrightarrow{\text{required}} 5\left(x + \frac{y}{4}\right)$

$\longrightarrow 5x$

i.e. $5x = 15$ ($\because$ Given 15 ml of CO_2)

$x = 3$

$$5\left(x + \frac{y}{4}\right) = 20$$

$y = 4$

M.F of hydrocarbon = C_3H_4